

RasptotV2 – Tomato detection and classification in a greenhouse using lightweight YOLOv8s-seg and computer-vision rules

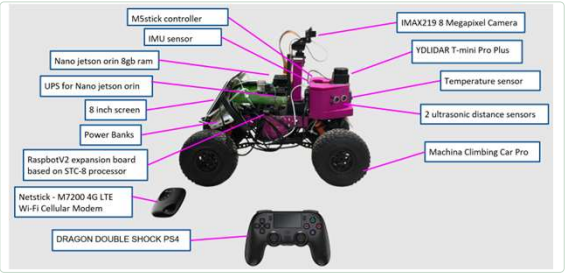
Final Project – Stage B | Mobile robot in a crowded greenhouse scene | Jetson Orin Nano, ROS2 Humble, LIDAR, camera and environmental sensors

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Project code: 26-1-R-6

The main research question

To what extent does combining a lightweight YOLOv8s-seg model with computer-vision rules based on color, masks, density, temporal consistency and cluster structure improve detection and classification of tomatoes and tomato clusters in crowded greenhouse scenes, while maintaining real-time operation on a resource-limited robotic system?

Experimental environment and robot



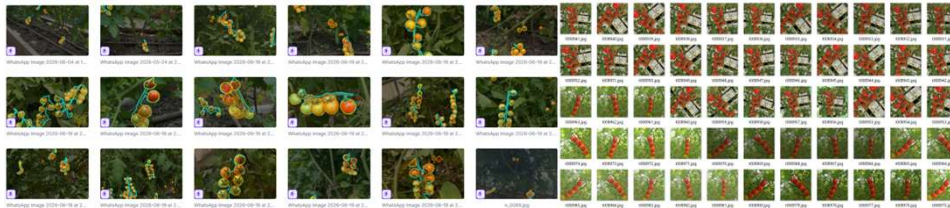
Jetson Orin Nano 8GB runs inference and rule-based decisions.

LIDAR + ROS2 Humble SLAM for real-time map building.

M5Stick + IMU + environmental sensors for greenhouse sampling.

IMX219 8MP camera, ~75° FOV: relatively weak, yet sufficient for an initial real-time greenhouse sampling.

Dataset, training and hybrid detection pipeline



Manual transmission 446 segmentation images in ROBOFLOW and only 20 more images from the hotbed itself, some of which were in a wide image like the one the robot sees while driving.

Image augmentation: angles, lighting, and crowded scenes. After all the image augmentation, the training set increased to 34,330

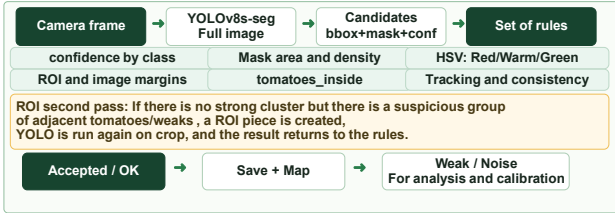
Lightweight model: YOLOv8s-seg for real-time recognition + segmentation mask.

Classes: ripe, unripe, ripe bunch, unripe bunch:

Output: bbox, mask, confidence and color within the mask.

Idea: Small model + rules  
To work on a robot with limited resources.

Hybrid Forecast Chart



Ripe cluster - strong identification

Identifying a ripe bunch on a red cluster. The rules use warm/red color, masking, and support of ripe tomatoes to confirm the cluster.



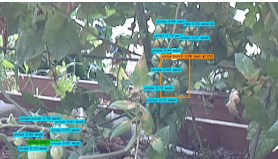
Ripe cluster - repair and strengthening

Example of a ripe cluster where the internal support of ripe detections strengthens the decision and helps to correct discrepancies with the unripe class



Unripe cluster - cluster promotion

Several adjacent green/yellow tomatoes create cluster support. Even if some of the predictions are weak, the spatial structure strengthens the decision.



Unripe cluster - internal support

The model detects individual tomatoes, and the rules check tomatoes\_inside, color, and mask density to confirm an unripe bunch



Weak focus - noise cleaning

Concentrating weak predictions around fruits could be used in the future for ROI weak density and temporal promotion, instead of treating every weak as noise.



Weak focus - 3 main areas

The image has 3 main regions of interest that explicitly show 3 clusters, some are also hidden, without leaf noise or false predictions. The strict regularity cleaned up the noise in real time.



Focus weak - 2 noise-free zones

The image has 2 main regions of interest without noise, with predictions that the model predicted strongly, but the rule set created inconsistencies that classified the objects as weak. In the following study, we will sort out the conflicts and return to resampling in the greenhouse.

What do the pictures highlight?

Clusters are the heart of the research: single tomatoes are detected well, but clusters require structural and color-based rules.

The cost:  
Hard thresholds may classify real objects as weak.  
Next direction: a soft score instead of pass/fail.

Weak ≠ Noise only:  
Adjacent weak hotspots can indicate a real cluster and serve for focused ROI, mask density and temporal support.

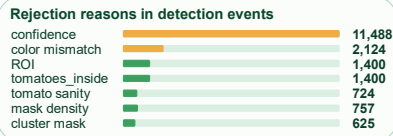
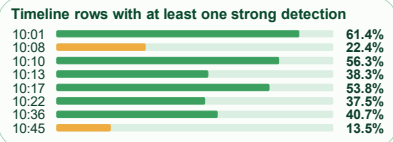
The rules clean noise: color, mask, ROI and tomato sanity reduce false predictions in a crowded environment.

Results across 8 greenhouse sessions

|                                                             |                           |
|-------------------------------------------------------------|---------------------------|
| YOLOv8s-seg + rule-based filtering on real greenhouse scans |                           |
| 8 sessions                                                  | 22.1 min total scan time  |
| 11,518 video frames                                         | 14,792 raw candidates     |
| 13,560 detections after rules                               | 1,367 accepted detections |
| 721 OK images                                               | 576 weak/noise images     |
| 123 cluster promotions                                      | 16 class corrections      |
| 3/61 accepted ROI second pass                               | 10.1% accepted rate       |

Class distribution

| class        | total | OK  | rate  |
|--------------|-------|-----|-------|
| unripe       | 9,619 | 631 | 6.6%  |
| ripe         | 2,161 | 561 | 26.0% |
| unripe bunch | 1,689 | 148 | 8.8%  |
| eripe bunch  | 91    | 27  | 29.7% |



Session summary

| time  | OK  | weak | acc | cluster |
|-------|-----|------|-----|---------|
| 10:01 | 203 | 95   | 438 | 28      |
| 10:08 | 43  | 50   | 68  | 8       |
| 10:10 | 73  | 41   | 194 | 23      |
| 10:13 | 50  | 50   | 90  | 3       |
| 10:17 | 128 | 92   | 224 | 32      |
| 10:22 | 75  | 72   | 111 | 16      |
| 10:36 | 129 | 110  | 218 | 15      |
| 10:45 | 20  | 66   | 24  | 3       |

Conclusions and future work

Key result: a lightweight YOLOv8s-seg model can deliver real-time prediction in a crowded greenhouse scene when a rule layer is added on top of the model output.

Contribution of the rules: color sanity, mask, density and tomatoes\_inside removed many false predictions and made the system more conservative and reliable.

Clusters: cluster promotion and class correction were the most significant rules for identifying ripe and unripe clusters.

Weak detections: not every weak is noise. Adjacent weak hotspots can indicate a region with a model and serve for ROI/temporal promotion.

Camera limitation: the system works with an IMX219 8MP camera with ~75° FOV, a relatively simple and weak camera. Nevertheless, good results were obtained for initial real-time sampling in the greenhouse.

Open problem: hard thresholds, especially confidence, may reject real candidates. The next direction is a soft, weighted score.

Future research: calibrate the rules based on the 8 sessions, leverage weak heatmaps, perform resampling and measure precision/recall against manual ground truth.

The poster's message:

The rules do not replace the model, they turn the lightweight model output into useful engineering information: acceptance, rejection, cluster reinforcement, class correction and identification of points of interest for further sampling.